Fiscal Year:	FY 2010	**Task Last Updated:**	FY 04/12/2010
PI Name:	Thiessen, David Brian Ph.D.		
Project Title:	Phase Separation and Phase Change Using Helically-Supported Capillary Channels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	thiessen@wsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	509-335-4908
Organization Name:	Washington State University		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	1405 Northeast College Ave, MS 2814		
PI Web Page:			
City:	Pullman	**State:**	WA
Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/31/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Green, Robert D	**Contact Phone:**	216-433-5402
Contact Email:	robert.d.green@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marston, Philip (Washington State University)		
Grant/Contract No.:	NNX09AI82G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Wicking structures with low resistance to axial flow are currently used in heat pipes and may have other applications in microgravity associated with phase separation and phase change technologies. The proposed work involves a fundamental study of a novel capillary flow channel consisting of a helical-wire support that is potentially useful in technologies and benefits from having a symmetry that simplifies the analysis of stability and flow dynamics. The channel geometry makes it particularly easy to measure the capillary dynamics on a global scale using high-speed imaging and image analysis. It is proposed to study the capture of droplets from a flowing air stream by a liquid-filled channel and the capture of bubbles from a flowing liquid stream by a gas-filled channel. Phenomena to be understood include the drop or bubble coalescence event and the subsequent response of the channel as capillary waves propagate from the impact zone dissipating energy. A model for capillary wave dynamics and for dynamic flows in these channels will be developed. Models for drop and bubble capture will also be developed that will be coupled to the channel dynamics model. Studies of phase change at an extended meniscus supported by the helical wire are also proposed. In		

	all cases the basic component to be studied is the capillary channel supported by a helical wire and connected to a constant-pressure reservoir. The behavior of multichannel arrays for phase separation will also be investigated. Parabolic flight experiments are planned and ISS facilities are identified for possible future experiments. The understanding gained from this project is expected to be applicable to other channels that are based on the same fundamental physics. The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices.
Task Progress:	Progress has been made on two research objectives. First we have made progress on experimental studies to elucidate the dynamics of drop capture by a liquid-filled channel. High-speed imaging of individual drop-capture events has been accomplished using a unique experimental configuration that allows for very detailed analysis of the dynamics from two orthogonal points of view, axial and lateral. The axial imaging is most useful for getting detailed information on the drop dynamics during and shortly after initial impact. Lateral imaging is most useful for getting details on the dynamic channel response following drop impact. Image analysis methods have been developed to track the dynamic channel diameter response. These methods have allowed us to detect pressure waves that rapidly run back and forth on the channel for a fraction of a second followed by a slower diameter relaxation during which the mass of the absorbed drop is pumped by capillarity to the pressure reservoir. The drop-capture apparatus has also been used to measure the capture efficiency as a function how far off-center the drop impacts the channel. For low impact speeds (< 1 m/s) and centered impact, the mass fraction of impacting drops absorbed by the channel can be around 80%. The effects of pitch, impact speed, and channel pressure have also been explored with preliminary measurements. The second objective on which progress has been made is on the development of a model for capillary wave dynamics and for dynamic flows in the capillary channels. A one-dimensional model has been developed based on cross-section-averaged momentum and mass conservation. This model predicts wave speeds that are in qualitative agreement with observation. A detailed theoretical study of capillary waves on a liquid cylinder with an array of equally spaced concentric rings that pin the surface has also been accomplished. An evanescent wave expansion method is used for wave scattering by a single ring and then a wide-spacing approximation allows one to extend those results to an array of multiple rings. This solution has been found to agree with a finite element model. A solution that takes advantage of the Bloch theorem has also been produced to predict wave transmission on a liquid cylinder with an infinite array of rings. The theoretical capillary-wave studies are expected to be useful for refining some aspects of the one-dimensional model for channel dynamics. An understanding of channel dynamics is expected to be important in understanding how the drop-capture efficiency depends on channel properties.
Bibliography Type:	Description: (Last Updated: 09/14/2018)
Abstracts for Journals and Proceedings	Thiessen D, Oelerich J, Cline K, Zhang L. "Hydrostatics, steady flow, and dynamics in helically-supported capillary channels." Presented at the APS 62nd Annual Meeting of the Division of Fluid Dynamics, Minneapolis, MN, November 22-24, 2009. Bulletin of the American Physical Society 2009 Nov;54(19):BG3. , Nov-2009